

Dan Vu

Electrical engineering professional with experience in signal processing, research automation, and sensor devices

danvu772@gmail.com • (808) 276-9135 • linkedin.com/in/danvu772

Education

University of Hawaii at Manoa

2021 – 2026

Bachelor of Science, Electrical Engineering – Systems and Data Science

GPA: 3.70

Projects & Work Experience

Ray's Research Group – Biomedical Sensor Development Lab

Honolulu, HI

Undergraduate Research Assistant

09/2024 – Present

- Built a Python-based automated test system for sensors, reducing test time from hours to instant processing
- Engineered a multi-threaded real-time data acquisition pipeline synchronizing multirate data sources
- Built analysis pipeline across multi-batch sensor datasets to characterize sensor response and perform calibration
- Developed a minimal C++ IIR/FIR filter library to provide lightweight DSP functionality, supporting signal conditioning pipelines

Hawaiian Electric Company

Kahului, Hawaii

Relay Test Engineer Intern

06/2025 – 08/2025

- Automated relay test data workflows in Excel/VBA, reducing manual processing time by 3+ hours per task
- Maintained and expanded relay device database, adding 147 devices and correcting 150+ outdated records
- Supported Maui wildfire mitigation infrastructure by leading SCADA logic validation

Fast Enterprises, LLC – Hawaii Tax Modernization System

Honolulu, Hawaii

Implementation Consultant Intern

05/2024 – 08/2024

- Rebuilt production internal testing tool, enabling automated generation of diverse test datasets and accelerating QA cycles for 27-engineer development team
- Authored complex SQL queries and resolved critical reporting defects in live tax system, ensuring accurate extraction of business-critical financial data
- Delivered technical demo and led training session for 27 engineers, driving organization-wide adoption of new automated testing workflow

Honolulu Authority for Rapid Transportation (HART)

Honolulu, Hawaii

Core Systems Student Aide

06/2023 – 05/2024

- Developed a MATLAB and Python-based Ride Quality Assessment (RQA) tool using FFT spectrum analysis to quantify train vibrations against ISO benchmarks, eliminating \$2,500-\$3,000 per test in outsourced costs
- Automated multi-axis accelerometer data analysis pipeline, reducing manual post-processing time by 50%
- Produced technical reports used by city, county, and contractor teams to inform operational decisions
- Performed track testing and developed track measurement correction workflow

Smart Campus Energy Laboratory – Distributed Weather Sensor Network

Honolulu, Hawaii

Firmware Team Member

09/2023 – 05/2024

- Maintained 24/7 multi-sensor weatherbox network across 3 deployed field devices
- Troubleshoot and recommissioned faulty sensors including component replacement and firmware verification
- Migrated legacy Python 2 data pipelines to Python 3, resolving compatibility failures in live sensor streams
- Repaired database schema issues to restore structured real-time sensor data storage

Skills

Embedded & Instrumentation: Real-time data acquisition (DAQ), digital filter implementation, Analog signal conditioning, Oscilloscopes, Waveform generators, Arduino/ATmega MCUs, UART, PyMeasure, scipy

Signal Processing & Analysis: DSP (filtering, spectral analysis, compression) FFT, SciPy signal, MATLAB, Sensor calibration and response characterization

Software & Data: Python (NumPy, Pandas, SciPy, Matplotlib), C++, C, SQL, Flask, Celery, Git

Languages: English (Native), Vietnamese (Professional fluency)